

## Task 1: Data Pipeline for LoRaWAN Sensor Insights

**Description:** You are working with uplink sensor data collected from 5,000+ IoT devices using LoRaWAN technology. The data is available in a CSV format and contains information like signal strength, temperature, humidity, GPS location, etc. You are required to ingest the data into MongoDB and run a few analysis queries to extract insights.

**Dataset:** lorawan\_uplink\_devices.csv

### Task Details:

1. Write Python script to import the CSV into a MongoDB collection named uplinks.
2. Identify the 10 devices with the highest number of uplinks.
3. Find the average RSSI and SNR for each device. Sort by the lowest RSSI to detect weak connections.
4. Compute the average temperature and humidity for each gateway\_id.
5. Find all device\_ids that have more than one record (i.e., duplicate devices).
6. Export a JSON list of all records where temperature is above 35°C, including only device\_id, latitude, longitude, and temperature.
7. Scheduling the script to run daily for new devices addition. **(Bonus Point)**

### Evaluation Criteria:

- Correctness of functionality.
- Code structure, readability, and use of best practices.
- Error handling, Logging and edge case management.
- Understanding of IoT device identifiers and data flow.

## Task 2: MongoDB Aggregation

**Description:** You are given a MongoDB collection that contains e-commerce sales data. Write aggregation queries to answer the following business analytics tasks using **Python** and **PyMongo**. Recommended to use MongoDB Aggregation.

**Dataset:** orders.csv

### Task Details:

1. Find the top 5 products (Product ID) with the highest total Sales.
2. Calculate the total revenue (Sales) generated each month across all years (Order Date).
3. Group orders by Category and calculate the average sales amount per Sub-Category.
4. Calculate the total sales for each year (Order Date) and show percentage growth from the previous year.